Infinite Connections

Generating New NYT-Style Word Puzzles at Scale

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The problem

The New York Times Connections puzzle shows the player 16 words and asks them to sort the words into 4 groups of 4 based on a hidden theme. The themes can be simple (four synonyms for "hello"), tricky (four things that fit before the word "fish"), or outright playful (four Taylor Swift album titles). For this project we built a system that CREATES these puzzles automatically, at scale, with a web app that lets anyone play them in the browser.

Generating a good Connections puzzle is harder than it looks. The 16 words must fit together in exactly one way -- no ambiguity. The themes must feel fresh but recognizable. And we cannot accidentally reproduce a puzzle the NYT has already published, which the grading rubric treats as an automatic failure.

What we built -- five complementary methods

Rather than pick one approach and hope it works, each teammate built a different method. Together they cover the full design space from "ask the AI everything" to "use no AI at all."

- **Adreama -- Iterative AI generation.**
  Asks the AI to build one group at a time, remembering what it has already used. Built up a library of 400+ puzzles with memory that persists across sessions, so duplicates never creep in even across many runs.
- **Burak -- Pattern-driven dictionary builder.**
  Uses no AI for generation. Instead, pulls words from language databases (WordNet, ConceptNet, Datamuse) and fills puzzle slots using explicit patterns (synonyms, category membership, shared actions). Specializes in the green and blue difficulty levels with thousands of candidate groups.
- **Hengkai -- Quality-first yellow specialist.**
  Focused on the easiest (yellow) difficulty, where cohesion and obviousness matter most. Uses cached word embeddings and strict filters that throw out any word too obscure or technical.
- **Kaiwen -- Two pipelines, compared head to head.**
  Pipeline A follows the published 'Making New Connections' paper: five AI calls per puzzle. Pipeline B is our new method, CATEGORY-FIRST RETRIEVAL (CFR): the AI proposes only the category names (one call per puzzle, or zero if we reuse NYT categories), then math finds the matching words from an expanded 14,877-word dictionary.
- **Abuzar -- Multi-agent critic system.**
  Builds puzzles with one AI agent and has a second agent critique the result. Good ideas survive, bad ones get rewritten. Pairs this with a playable web server and a library of 'concept inspirations' to keep themes fresh.

All five methods share the same SAFETY CHECKS: every candidate puzzle is compared against the 554 real NYT puzzles both as a whole 16-word set AND group-by-group. If any generated group exactly matches a real NYT group, we throw that group out and try again. This means we cannot fail the 'don't copy the NYT' rule -- the system refuses to emit anything that would trigger it.

Results

Across all methods we generated roughly 2,000+ validated puzzles. The best pipeline (CFR) achieves a 99% pass rate on automated validation, reproduces ZERO past NYT puzzles, and uses words from a 14,877-word vocabulary -- three times larger than the NYT original. At full scale, generating 10,000 puzzles costs between $0 and $50 depending on the method, so we can produce as many as the course requires.

What we learned

The surprising finding was that the simplest, least-AI-heavy method produced the highest pass rate. Asking the AI to do everything (pick words, pick categories, pick difficulty) is both expensive and unreliable: the AI gives inconsistent word choices and needs a separate editor pass to clean up. But letting a language AI do only THE CREATIVE NAMING step -- and letting math handle the mechanical word-lookup step -- produces better results at one-twentieth the cost and four times the speed.

Across the team, the strongest puzzles tended to come from methods that combined an AI's creativity with deterministic math for selection: everyone's pipeline used embeddings (a way to compare how similar two words are numerically) somewhere in the process.

Deliverables

- **Live web app**
  kevin2330.github.io/nyt-connections-generator/ loads 540+ validated puzzles in the NYT Connections interface. Anyone can play them in a browser.
- **Master Jupyter notebook**
  notebooks/master_demo.ipynb walks through every method end-to-end in reproducible form. It runs without any API keys -- all AI calls fall back to realistic mock responses when none are configured.
- **Technical report**
  Methods document and per-teammate folders detail the algorithms, math, and benchmarks.
- **Pre-generated dataset**
  A large set of validated puzzles on disk, from which the course can sample for TA evaluation.

Future work

The next step is HUMAN evaluation, not just the automated solvers: put the puzzles in front of real puzzle players and measure how many they would believe came from the NYT. We added a mechanism for that to the web app (thumbs-up / thumbs-down on each puzzle) but haven't yet collected a large sample. A stretch goal is a fine-tuned embedding model trained specifically on Connections puzzles, which would likely push the pass rate even higher for puzzles built around wordplay (the one category where our current system is weakest).